Solid Geometry & Similarity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a rectangular prism with length 8, width 5, and height 6.

$V = lwh = 8 \times 5 \times 6 = 240$ cubic units.

2 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cylinder with radius 5 cm and height 14 cm, in terms of π.

$V = \pi r^2 h = \pi(25)(14) = 350\pi$ cm^3.

3 [Foundational] • Target: Volume of prisms and cylinders

A triangular prism has a triangular cross-section of area 24 cm^2 and length 15 cm. Find its volume.

$V = (\text{cross} - \text{section area}) \times \text{length} = 24 \times 15 = 360$ cm^3.

4 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a cube with side length 9.

$V = 9^3 = 729$ cubic units.

5 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a square pyramid with base side 10 and height 12.

$V = \frac{1}{3}(\text{base area})(\text{height}) = \frac{1}{3}(100)(12) = 400$ cubic units.

6 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a cone with radius 6 and height 15, in terms of π.

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(15) = 180\pi$ cubic units.

7 [Foundational] • Target: Volume of pyramids and cones

A cone has volume 300π cubic units and height 25. Find its radius.

$300\pi = \frac{1}{3}\pi r^2(25) \Rightarrow r^2 = \frac{900}{25} = 36 \Rightarrow r = 6$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Volume of pyramids and cones

A rectangular pyramid has a base 6 by 8 and height 9. Find its volume.

$V = \frac{1}{3}(6 \times 8)(9) = \frac{1}{3}(48)(9) = 144$ cubic units.

9 [Intermediate] • Target: Volume of spheres

Find the volume of a sphere with radius 9, in terms of π.

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(729) = 972\pi$ cubic units.

10 [Intermediate] • Target: Volume of spheres

A hemisphere has radius 6. Find its volume, in terms of π.

$V = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi(216) = 144\pi$ cubic units.

11 [Intermediate] • Target: Volume of spheres

A spherical tank has diameter 10. Find its volume, in terms of π.

Radius $= 5$. $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3}$ cubic units.

12 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 3, in terms of π.

$V = \frac{4}{3}\pi(27) = 36\pi$ cubic units. $SA = 4\pi(9) = 36\pi$ square units.

13 [Intermediate] • Target: Surface area of solids

Find the surface area of a cylinder with radius 4 and height 10, in terms of π.

$SA = 2\pi r^2 + 2\pi rh = 2\pi(16) + 2\pi(4)(10) = 32\pi + 80\pi = 112\pi$.

14 [Intermediate] • Target: Surface area of solids

Find the surface area of a cone with radius 5 and slant height 13, in terms of π.

$SA = \pi r^2 + \pi rl = 25\pi + 65\pi = 90\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Surface area of solids • Cross-Lesson: 8.01 Statistics & Probability

Find the surface area of a sphere with radius 7, in terms of π.

$SA = 4\pi r^2 = 4\pi(49) = 196\pi$.

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16 [Challenging] • Target: Surface area of solids • Cross-Lesson: 1.01 Advanced Algebra

Find the surface area of a cube with side length 7.

$SA = 6(7^2) = 6(49) = 294$ square units.

17 [Challenging] • Target: Composite solids • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A solid consists of a cylinder (radius 3, height 10) topped by a cone with the same radius and height 4. Find the total volume, in terms of π.

Cylinder: $\pi(9)(10) = 90\pi$. Cone: $\frac{1}{3}\pi(9)(4) = 12\pi$. Total $= 90\pi + 12\pi = 102\pi$.

18 [Challenging] • Target: Composite solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A rectangular box measuring 8 by 6 by 4 has a cylindrical hole (radius 2, drilled fully through the height of 4) removed. Find the remaining volume, in terms of π.

Box volume $= 8 \times 6 \times 4 = 192$. Cylinder removed $= \pi(4)(4) = 16\pi$. Remaining volume $= 192 - 16\pi$.

19 [Challenging] • Target: Composite solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

A solid consists of a cylinder (radius 4, height 8) topped by a hemisphere of the same radius. Find the total volume, in terms of π.

Cylinder: $\pi(16)(8) = 128\pi$. Hemisphere: $\frac{2}{3}\pi(64) = \frac{128\pi}{3}$. Total $= 128\pi + \frac{128\pi}{3} = \frac{512\pi}{3}$.

20 [Challenging] • Target: Composite solids • Cross-Lesson: 5.01 Trigonometry

A solid is a rectangular box (10 by 6 by 5) with a square pyramid (base 6 by 6, height 4) carved out of the top. Find the remaining volume.

Box: $10 \times 6 \times 5 = 300$. Pyramid: $\frac{1}{3}(36)(4) = 48$. Remaining volume $= 300 - 48 = 252$ cubic units.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Volume of prisms and cylinders

Find the volume of a rectangular prism with length 14, width 10, and height 13.

a. 1820 ✓

b. 1824

c. 37

d. 1816

$V = lwh = 14 \times 10 \times 13 = 1820$ cubic units.

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22 [Foundational] • Target: Volume of pyramids and cones

Find the volume of a rectangular prism with length 5, width 7, and height 10.

a. 350 ✓

b. 346

c. 354

d. 22

$V = lwh = 5 \times 7 \times 10 = 350$ cubic units.

23 [Foundational] • Target: Volume of spheres

Find the volume of a rectangular prism with length 13, width 11, and height 11.

a. 1569

b. 35

c. 1573 ✓

d. 1577

$V = lwh = 13 \times 11 \times 11 = 1573$ cubic units.

24 [Foundational] • Target: Surface area of solids

Find the volume of a rectangular prism with length 8, width 7, and height 13.

a. 28

b. 724

c. 728 ✓

d. 732

$V = lwh = 8 \times 7 \times 13 = 728$ cubic units.

25 [Foundational] • Target: Composite solids

Find the volume of a rectangular prism with length 12, width 3, and height 10.

a. 356

b. 364

c. 360 ✓

d. 25

$V = lwh = 12 \times 3 \times 10 = 360$ cubic units.

26 [Foundational] • Target: Similar solids

Find the volume of a rectangular prism with length 5, width 12, and height 4.

a. 236

b. 240 ✓

c. 244

d. 21

$V = lwh = 5 \times 12 \times 4 = 240$ cubic units.

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27 [Foundational] • Target: Cross-sections of solids

Find the volume of a rectangular prism with length 13, width 15, and height 10.

a. 1946

b. 1954

c. 38

d. 1950 ✓

$V = lwh = 13 \times 15 \times 10 = 1950$ cubic units.

28 [Foundational] • Target: A well-designed word problem: a grain silo

Find the volume of a cone with radius 6 and height 18, in terms of π.

a. 864π

b. 648π

c. 216π ✓

d. 218π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(18) = 216\pi$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Polyhedra & Euler's Formula

Find the volume of a cone with radius 3 and height 8, in terms of π.

a. 72π

b. 24π ✓

c. 26π

d. 96π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(8) = 24\pi$.

30 [Intermediate] • Target: Cones, Pyramids & Frustums

Find the volume of a cone with radius 4 and height 9, in terms of π.

a. 48π ✓

b. 50π

c. 192π

d. 144π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(16)(9) = 48\pi$.

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31 [Intermediate] • Target: Spheres, Cylinders & Cross-Sections

Find the volume of a cone with radius 3 and height 19, in terms of π.

a. 59π

b. 171π

c. 57π ✓

d. 228π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(19) = 57\pi$.

32 [Intermediate] • Target: Scale Factors & Volume Ratios

Find the volume of a cone with radius 5 and height 14, in terms of π.

a. $\frac{356}{3}\pi$

b. $\frac{350}{3}\pi$ ✓

c. 350π

d. $\frac{1400}{3}\pi$

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(25)(14) = \frac{350}{3}\pi$.

33 [Intermediate] • Target: Volume of prisms and cylinders

Find the volume of a cone with radius 5 and height 16, in terms of π.

a. $\frac{1600}{3}\pi$

b. $\frac{406}{3}\pi$

c. $\frac{400}{3}\pi$ ✓

d. 400π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(25)(16) = \frac{400}{3}\pi$.

34 [Intermediate] • Target: Volume of pyramids and cones

Find the volume of a cone with radius 8 and height 2, in terms of π.

a. $\frac{128}{3}\pi$ ✓

b. 128π

c. $\frac{512}{3}\pi$

d. $\frac{134}{3}\pi$

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(64)(2) = \frac{128}{3}\pi$.

35 [Intermediate] • Target: Volume of spheres

Find the volume and surface area of a sphere with radius 7, in terms of π.

a. $\frac{1384}{3}\pi, 196\pi$

b. $\frac{1372}{3}\pi, 204\pi$

c. $\frac{1372}{3}\pi, 196\pi$ ✓

d. $\frac{1360}{3}\pi, 188\pi$

$V = \frac{4}{3}\pi(343) = \frac{1372}{3}\pi$ cubic units. $SA = 4\pi(49) = 196\pi$ square units.

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36 [Intermediate] • Target: Surface area of solids

Find the volume and surface area of a sphere with radius 11, in terms of π.

a. $\frac{5336}{3}\pi, 484\pi$

b. $\frac{5312}{3}\pi, 476\pi$

c. $\frac{5324}{3}\pi, 484\pi$ ✓

d. $\frac{5324}{3}\pi, 492\pi$

$V = \frac{4}{3}\pi(1331) = \frac{5324}{3}\pi$ cubic units. $SA = 4\pi(121) = 484\pi$ square units.

37 [Intermediate] • Target: Composite solids

A solid consists of a cylinder (radius 6, height 7) topped by a cone with the same radius and height 7. Find the total volume, in terms of π.

a. 252π

b. 336π ✓

c. 338π

d. 334π

Cylinder: $\pi(36)(7) = 252\pi$. Cone: $\frac{1}{3}\pi(36)(7) = 84\pi$. Total $= 252\pi + 84\pi = 336\pi$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Similar solids • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

A solid consists of a cylinder (radius 8, height 12) topped by a cone with the same radius and height 2. Find the total volume, in terms of π.

a. $\frac{2438}{3}\pi$

b. 768π

c. $\frac{2432}{3}\pi$ ✓

d. $\frac{2426}{3}\pi$

Cylinder: $\pi(64)(12) = 768\pi$. Cone: $\frac{1}{3}\pi(64)(2) = \frac{128}{3}\pi$. Total $= 768\pi + \frac{128}{3}\pi = \frac{2432}{3}\pi$.

39 [Challenging] • Target: Cross-sections of solids • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Two similar cylinders have heights 7 and 8. Find the ratio of their volumes.

a. $\frac{512}{343}$ ✓

b. $\frac{343}{512}$

c. $\frac{64}{49}$

d. $\frac{8}{7}$

Scale factor $k = \frac{8}{7}$. Volume ratio $= k^3 = \frac{512}{343}$.

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40 [Challenging] • Target: A well-designed word problem: a grain silo • Cross-Lesson: 5.01 Trigonometry

Two similar cylinders have heights 2 and 5. Find the ratio of their volumes.

a. $\frac{8}{125}$

b. $\frac{25}{4}$

c. $\frac{125}{8}$ ✓

d. $\frac{5}{2}$

Scale factor $k = \frac{5}{2}$. Volume ratio $= k^3 = \frac{125}{8}$.

41 [Challenging] • Target: Polyhedra & Euler's Formula • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A convex polyhedron has 12 vertices and 30 edges. How many faces does it have?

a. 18

b. 20 ✓

c. 22

d. 24

By Euler's formula $V - E + F = 2$: $12 - 30 + F = 2 \Rightarrow -18 + F = 2 \Rightarrow F = 20$ (an icosahedron).

42 [Challenging] • Target: Polyhedra & Euler's Formula • Cross-Lesson: 8.01 Statistics & Probability

What is the length of the space diagonal of a rectangular prism with dimensions $3, 4,$ and 12?

a. 13 ✓

b. 15

c. $\sqrt{160}$

d. 14

$d = \sqrt{l^2 + w^2 + h^2} = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

43 [Challenging] • Target: Cones, Pyramids & Frustums • Cross-Lesson: 1.01 Advanced Algebra

What is the volume of the frustum formed in the above cone with bases of radii 6 cm and 4 cm and height 3 cm?

a. 76π **cm**^{3} ✓

b. 72π cm^3

c. 84π cm^3

d. 96π cm^3

Volume of frustum: $\frac{1}{3}\pi h(R^2 + Rr + r^2) = \frac{1}{3}\pi(3)(36 + 24 + 16) = 76\pi$ cm^3.

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44 [Challenging] • Target: Spheres, Cylinders & Cross-Sections • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A sphere is intersected by a plane 5 cm from its center. If the radius of the cross-sectional circle is 12 cm, what is the radius of the sphere?

a. 13 **cm** ✓

b. 15 cm

c. $\sqrt{119}$ cm

d. 17 cm

$R = \sqrt{d^2 + r_{\text{circle}}^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm.

45 [Challenging] • Target: Scale Factors & Volume Ratios • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Two similar solid statues have heights in the ratio $2:5$. If the smaller statue weighs 16 kg, what does the larger statue weigh (assuming identical material)?

a. 250 **kg** ✓

b. 100 kg

c. 200 kg

d. 320 kg

Volume and weight scale with the cube of the linear factor: $(\frac{5}{2})^3 = \frac{125}{8}$. Weight $= 16 \times \frac{125}{8} = 250$ kg.

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